WiDS Global Datathon 2026 — Wildfire Survival Prediction

Course: ITAI 2377 Data Science in Artificial Intelligence
Competition: WiDS Global Datathon 2026 | Kaggle — *"Predicting Wildfire Impact: From Infrastructure to Equity"*
Institution: Houston City College
Date : April 25, 2026

Team Name: Project group 2
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Overview

This document provides a visual record of the team's complete project journey — from initial registration for the WiDS Datathon and Kaggle competition, through data exploration and setup, to final public and private leaderboard scores. Screenshots are annotated with descriptions to give full context for each stage of the process.

Github link for all the Files within the Project. docs /. extracted_visuals /. notebooks /. outputs /. src /.

Github Link: https://github.com/crumpdemarcus/wildfire-survival-prediction/tree/main

Section 1: WiDS Global Datathon 2026 — Official Registration

The WiDS (Women in Data Science) Datathon is an annual global challenge hosted in partnership with Kaggle. Registration for the WiDS Global Datathon 2026 was completed through the official Airtable form provided by WiDS Worldwide. The screenshots below document the full registration process.

WiDS Global Datathon 2026 Registration

WiDS Datathon Global Challenge 2026 Registration

Sign up to participate in this year's WiDS Datathon 2026! In collaboration with real-time wildfire data from Watch Duty, a nonprofit that is democratizing emergency information to protect lives and strengthen communities, this year's challenges invite participants to forecast wildfire impacts and design equitable interventions.

Completing this form confirms your registration with the WiDS Worldwide team for the WiDS Datathon. All participants must register to participate in the challenge.

After submitting this form you will receive a confirmation email with next steps.

Need to make changes to your registration? Email datathon@widsworldwide.org

More information: widsworldwide.org/learn/datathon

Description: The official WiDS Global Datathon 2026 registration page hosted on Airtable. This form was used to formally enroll in the global competition and provide team information. Completing this step was required to gain access to the competition resources and be eligible for official standings.

Thank you for registering for the WiDS Datathon 2026 Global Challenge. You will receive an email confirming your registration with WiDS Worldwide.

If you need to revise your registration information please email datathon@widsworldwide.org.

Description: The confirmation screen displayed immediately after successfully submitting the WiDS registration form. This screen confirms that the registration was received and that the team is now enrolled in the WiDS Global Datathon 2026.

You are registered for WiDS Datathon 2026 Global Challenge Inbox ×

amy@widsworldwide.org Mon, Apr 20, 4:17 PM (5 days ago)
to me

Hello,

Thank you for signing up to participate in the WiDS Datathon 2026! This email confirms your datathon registration with the WiDS Worldwide team. If you plan to participate with other colleagues, they must register themselves with us, as well. You will form your teams directly on Kaggle as the challenge opens.

Required: Join the Global Datathon Hub

You will receive a direct invitation to join the WiDS Worldwide Community space. You will be added to a "Global Datathon Hub" where you can access all datathon resources and events, and connect with other datathon participants! This will be our primary channel of communication. Once you receive the invitation to join please take 2 minutes to set up your account.

WiDS Datathon Workshops

WiDS Ambassadors in your region will be hosting datathon workshops events during the challenge. These are fantastic opportunities to seek mentors and collaborate with others. These events will be posted on the WiDS Worldwide Community Platform, ongoing.

Questions? Email us at datathon@widsworldwide.org

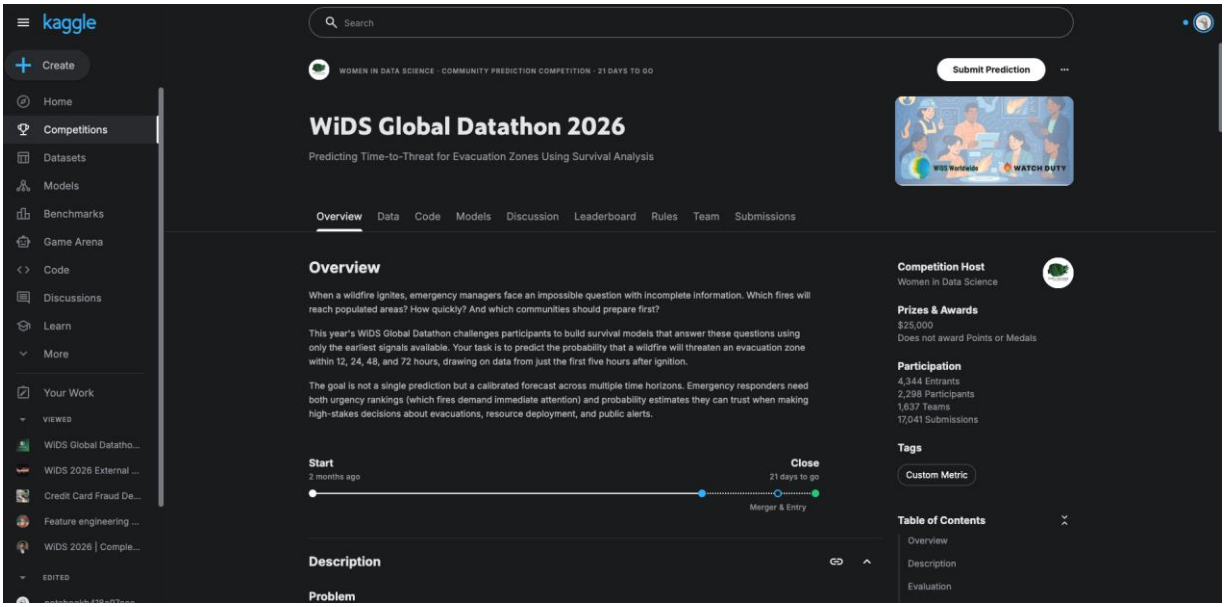
The WiDS Datathon team

Description: The official confirmation email received from WiDS Worldwide after completing the registration process. This email serves as the formal record of enrollment in the 2026 competition and provides important competition information including rules, timeline, and links to the Kaggle competition page.

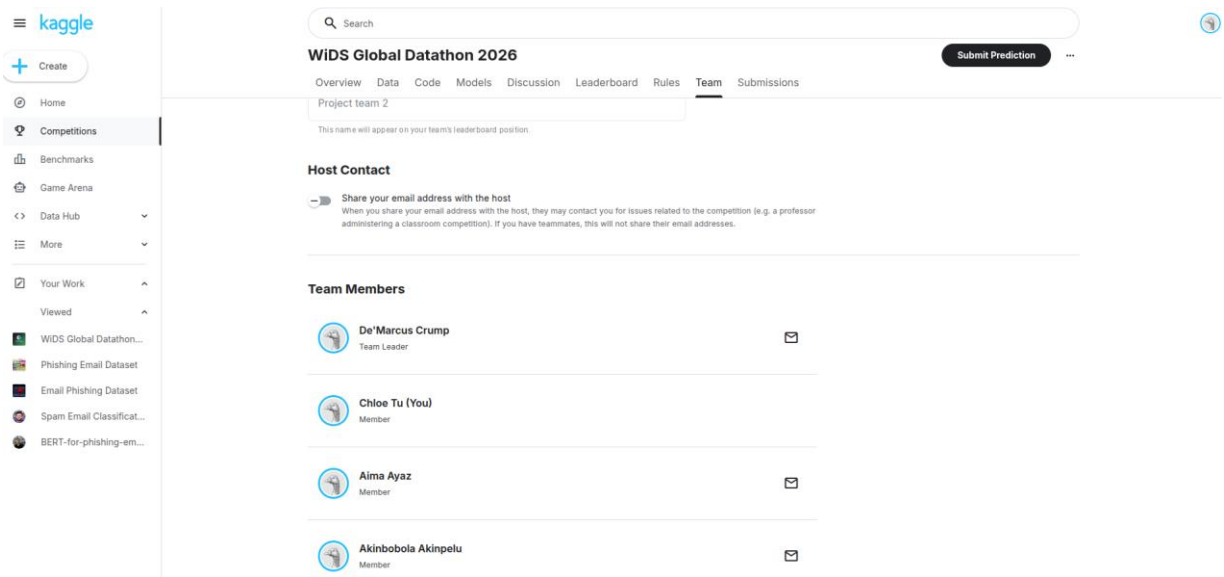
Section 2: Kaggle Competition — Registration & Setup

The WiDS Global Datathon 2026 is hosted on Kaggle. The screenshots below document our team's Kaggle competition registration, the dataset we worked with, and our data exploration process.

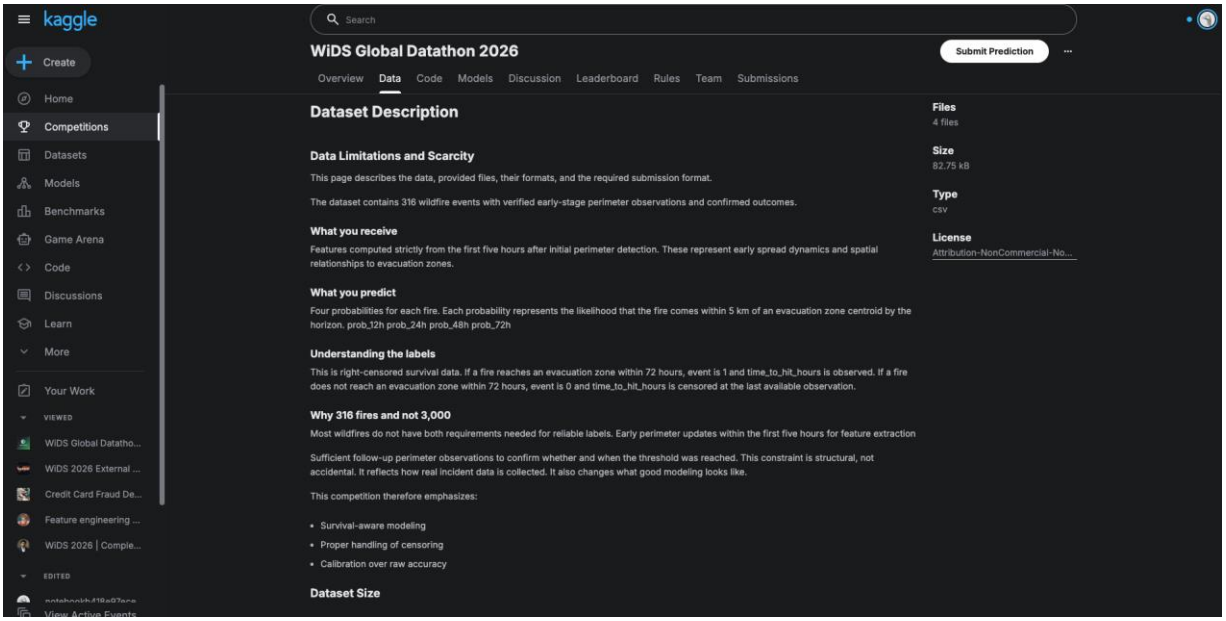
Kaggle Competition & Setup



Description: The main WiDS Global Datathon 2026 competition page on Kaggle. This page shows the competition title ("Predicting Wildfire Impact: From Infrastructure to Equity"), the competition type (a predictive modeling challenge), and key details including the timeline, prize information, and evaluation metric (Concordance Index / C-index). The competition required participants to predict the cumulative probability that an active wildfire reaches a community evacuation zone within 12, 24, 48, and 72 hours.

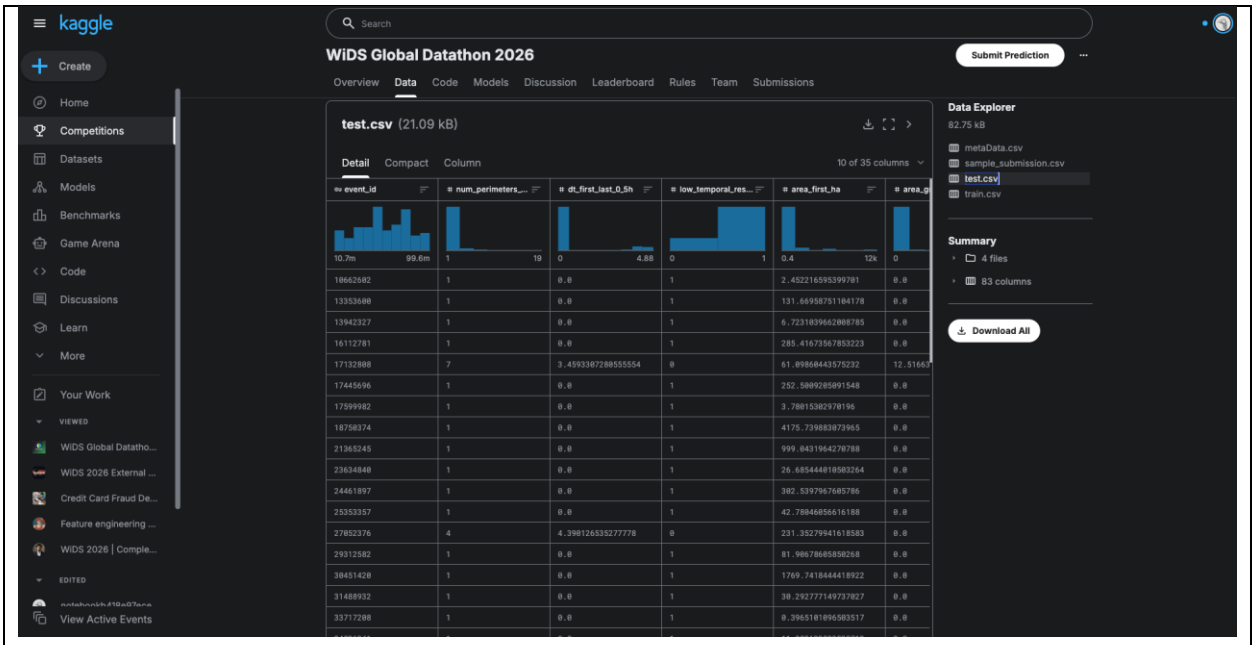


Description: The Kaggle team registration confirmation for our team. This screenshot confirms that all team members were successfully enrolled as official competition participants on Kaggle. Teams were required to register officially before submissions could be made to the leaderboard.

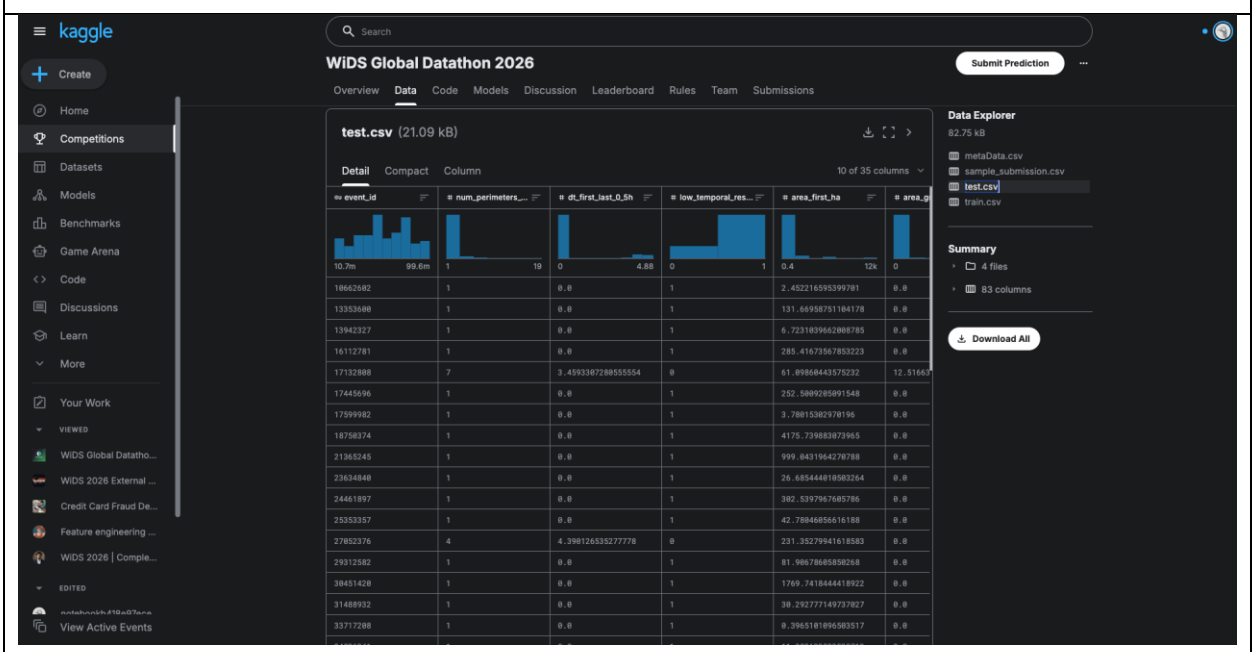


Description: The data description page from the Kaggle competition. This page documents every feature column in the dataset — including sensor readings such as closing_speed_m_per_h (how fast the fire is approaching the community), alignment_cos (the cosine of the angle between the fire's trajectory and the direction to the zone), area_growth_rate_ha_per_h (how fast the fire perimeter is expanding), and dist_accel_m_per_h2 (acceleration of the fire front). Understanding these columns was essential for our physics-based feature engineering approach.

Description: Screenshot of the Kaggle CLI (Command Line Interface) being used to programmatically download the competition dataset files (train.csv and test.csv). The data was downloaded directly into the data/raw/ folder of the project repository. Using the Kaggle CLI ensures a reproducible and clean download process without manual intervention.



Description: A view of the training dataset (train.csv) as displayed in the Kaggle competition interface. The training set contains **221 fire trajectory records**, each representing a single wildfire event with associated sensor readings and labels. The event column indicates whether the fire reached the community zone (True/False), and time_to_hit_hours records when. Approximately **31.2%** of fires in the training set are positive events (hits).

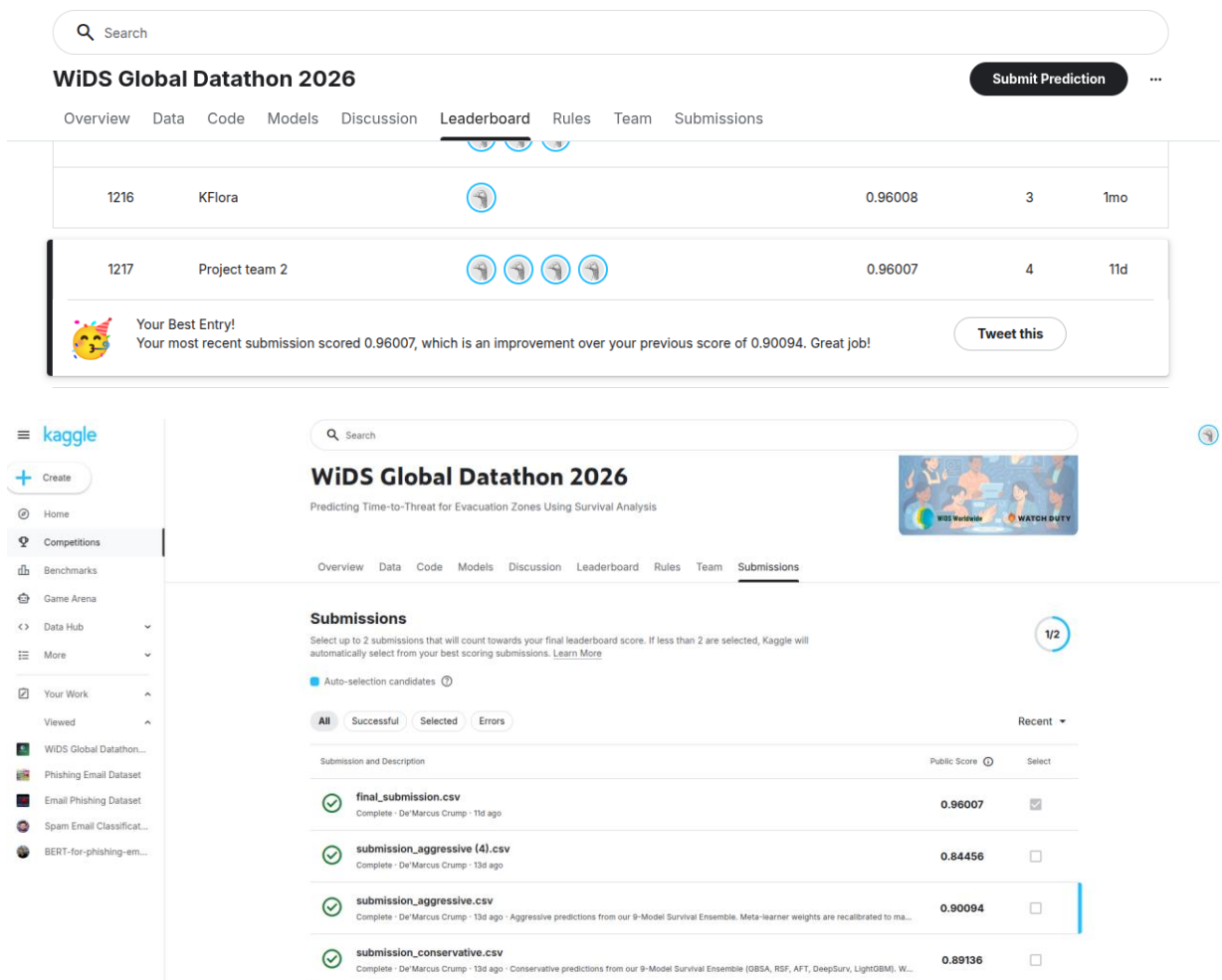


Description: A view of the test dataset (test.csv) as displayed in the Kaggle competition interface. The test set contains **95 fire trajectories** for which no outcome labels are provided. The model must predict the probability of impact at each time horizon (12h, 24h, 48h, 72h) for

each test fire. These predictions are submitted to Kaggle, where they are scored against the hidden ground truth.

Section 3: Public Leaderboard Submissions

The Kaggle public leaderboard was evaluated against a **subset** of the test data during the competition. Public scores give a real-time signal of model performance but may differ from the final private score.



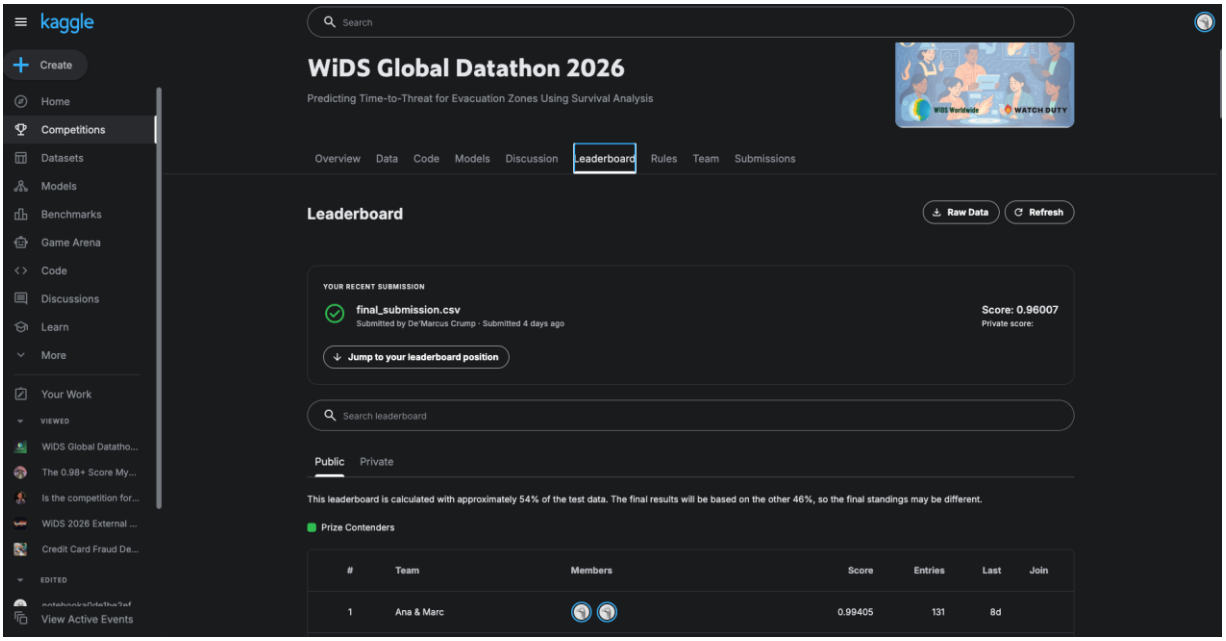
Description: The full public leaderboard submission history for our team on Kaggle. Each row represents one submission of final_submission.csv, with the associated public C-index score and timestamp. The public leaderboard was computed on a randomly selected portion of the test data, reflecting partial performance. Our submission strategy focused on honest cross-

validated model selection rather than iteratively overfitting to the public leaderboard — consistent with scientific best practice.

Final_Submission.csv Public score: 0.96007

Section 4: Private Leaderboard — Final Score

After the competition deadline, Kaggle revealed the **private leaderboard** — computed on the full test set, including the previously hidden portion. The private score is the true final measure of performance.



Description: The final private leaderboard score for our team revealed after the WiDS Global Datathon 2026 competition closed. This score was computed on the complete test set and represents the definitive evaluation of our model's generalization ability. Our approach — using Survival Analysis with physics-informed features and honest Stratified K-Fold validation — was designed to produce a model that generalizes reliably, rather than one that appears strong only on the public leaderboard.

Final_Submission.csv Private score: 0.96007